A Recurrent Narrow Ca II Emission Component in the Gaseous Debris Disk of WD J2100+2122

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Abstract

The white dwarf WD J2100+2122 hosts a variable gaseous debris disk whose Ca II triplet emission brightens episodically. In public VLT/X-shooter spectra from 2025 May 24 the excess emission is a single narrow component (FWHM ≈45 km s^{-1}, near the instrumental resolution) at -263 ± 2 km s^{-1} relative to the systemic velocity. A uniform reanalysis of all 29 public X-shooter nights from 2019 to 2025 identifies the same component, at velocities consistent within about 10 km s^{-1} across reduction choices, in the asymmetric 2022 May 24 high state and inside the broadly symmetric 2023 August 5 high state, but not distinctly in the strong double-peaked state of 2022 April 30. It appears in all three triplet lines and fades within days; its recurrence across three epochs suggests a persistent or repeatedly excited emission region, while the sparse sampling does not establish its geometry or constrain precession.

Keywords: White dwarf stars (1799) — Debris disks (363) — Circumstellar gas (238)

Context

WD J2100+2122 (Gaia DR3 1837948790953103232) is a hot DA white dwarf ($T_{\rm eff} \approx 25{,}600$ K, $\log g = 8.10$; Rogers et al. 2025) with a dust disk and a rich, variable emission-line gas disk (Dennihy et al. 2020; Melis et al. 2020). Dennihy et al. (2020) saw the triplet strengthen and fade within two months in 2019, through a blue-shifted asymmetric profile. Rogers et al. (2025) monitored the star with X-shooter and HAWK-I in 2022–2023 and found two gas brightenings coincident with K_s-band dust brightenings; the overall calcium profiles were asymmetric in the 2022 events and symmetric in 2023 August.

Data and measurement

All public X-shooter VIS spectra of the star (29 nights, 2019 July to 2025 May; Vernet et al. 2011) were retrieved as pipeline-reduced archive products. Nights before 2025 use the 0.9″ slit ($R \approx 8900$) or the 1.2″ slit ($R \approx 6500$). The 2025 May 24 pair (two 1485 s exposures, 1.2″ slit) was also re-reduced from raw frames with ESO X-shooter pipeline 3.8.3, each exposure separately, with a fixed spatial-profile extraction. Air wavelengths were barycentrically corrected and placed at the systemic velocity of -31.1 km s^{-1} (Rogers et al. 2025). Each spectrum was divided by a quadratic continuum fitted in six line-free intervals between 8450 and 8705 Å, resampled unsmoothed on a 0.2 Å grid matching the archive sampling, and differenced against the quiet 2023 September 20 spectrum. Figure 1 uses archive products throughout; 2025 numbers quoted below come from the raw re-reductions against the configuration-matched quiet 2023 May 2 pair unless stated. The primary 2025 comparison and all 2022–2023 comparisons are there-

fore between spectra of the same slit and resolution; cross-slit comparisons are checks, repeated with every night smoothed to $R = 6000$. Indices are integrals of the normalized difference over fixed windows: a narrow window at -327 to -207 km s^{-1} and a whole-line window of ± 12 Å. Pipeline-flagged pixels and a recurring sub-resolution sky residual at 8548.0 Å ($+239$ km s^{-1}) are excluded, not interpolated. Each index's noise floor is the standard deviation of the same integral at every line-free position of that night's difference spectrum, an empirical measure that includes between-night continuum and calibration mismatches. The narrow component was fitted with a Gaussian plus linear baseline over -450 to -90 km s^{-1}; errors come from 400 block-bootstrap refits of the residuals and exclude night-to-night wavelength zero-point errors, which may be a few km s^{-1}.

Results

Figure 1(a) shows the Ca II 8542 difference profiles of the four enhanced nights; the 2025 excess is a single narrow peak. In the separately re-reduced exposures its centroid is -261 ± 1 and -267 ± 2 km s^{-1}; in their mean the centroids are -267 ± 2, -263 ± 2, -267 ± 3 km s^{-1} at 8498, 8542 and 8662 Å, with FWHM 44–48 km s^{-1} against an instrumental 46 km s^{-1}, so the width is unresolved or nearly so. The narrow-window index at 8542 Å is 0.156 Å with a floor of 0.020 Å; the whole-line change is 0.36 Å. No excursion of the narrow-window integral anywhere in 261 Å of line-free continuum exceeds 0.07 Å. The archive product gives -264 ± 2 km s^{-1} and 0.150 Å against the September reference.

Both earlier high states show the same component. On 2022 May 24 the 8498, 8542 and 8662 centroids are -261 ± 2, -260 ± 3 and -257 ± 6 km s^{-1} (FWHM 42–68 km s^{-1}); on 2023 August 5 they are -264 ± 3, -264 ± 2 and -258 ± 5 km s^{-1} (FWHM 42–57 km s^{-1}), where it sits on a broad profile reaching ± 250 km s^{-1} whose overall shape is symmetric. These errors are statistical. Repeating the 8542 fits across two resolutions (unsmoothed and $R = 6000$), two quiet references and three baseline models (linear or quadratic over -450 to -90 km s^{-1}, linear over -380 to -150 km s^{-1}) gives centroids of -265 to -255, -266 to -259 and -264 to -263 km s^{-1} and widths of 38–74 km s^{-1}; this procedural spread of about 10 km s^{-1}, together with possible zero-point drift of a few km s^{-1}, sets the precision of the velocity agreement. Deconvolving the 34 km s^{-1} instrumental FWHM of the $0.9''$ nights leaves intrinsic widths of 25–59 km s^{-1} (30–44 at 8542 Å), subject to how well the instrumental profile is known. The strongest night, 2022 April 30, shows no comparably distinct narrow component: the same fits track the broad blue horn, returning -260 to -219 km s^{-1} and FWHM 40–123 km s^{-1}.

In Figure 1(b) the index exceeds 4σ on the four enhanced nights, where on 2022 April 30 the excess is the broad profile, and reaches 3.5σ (0.08 ± 0.02 Å) on 2022 May 26, two days after the 0.45 Å detection, tracing the decay; by May 31 it was 0.01 ± 0.03 Å, and 13 days after 2023 August 5 it was 0.01 ± 0.02 Å. The 24 other nights have a mean index of -0.009 Å with a sample standard deviation of 0.027 Å. The three detections differ in barycentric correction by 19 km s^{-1} and in slit and pipeline version, excluding a feature fixed in the telescope frame or on the detector. The mirror window at $+267$ km s^{-1} shows no counterpart in 2025 (0.02 ± 0.02 Å) and only the broad-component contribution in 2023 August. Between the two exposures of each detection night, 10 to 26 minutes apart, the 8542 centroid shifts by $+8.5 \pm 3.7$, 0.0 ± 3.6, -6.4 ± 2.7 km s^{-1}, changes comparable to possible zero-point drift.

Interpretation

The recurrence of a narrow, one-sided component at consistent projected velocities over three years indicates a persistent or repeatedly excited emission region. If the emitter is in circular Keplerian motion, $r/\sin^2 i \leq 1.9\,R_\odot$ for $M \approx 0.7\,M_\odot$, with equality at maximum projected velocity; the velocity lies within the range of the disk's double-peaked profile ($|v| \lesssim 390$ km s^{-1}; Rogers et al. 2025). Orbital motion of a localized emitter, an intermittently excited region at fixed azimuth, and a slowly precessing ec-

centric structure (precession timescales in such disks can exceed a century; Manser et al. 2016) are all compatible with the present cadence, which does not distinguish them. The data do not constrain the excitation mechanism or the relation to dust brightenings; the 2025 epoch has no simultaneous infrared photometry. Consecutive-night spectroscopy with simultaneous near-infrared photometry during a brightening would measure the component's rise and decay and its relation to dust production.

Data, acknowledgments and disclosure

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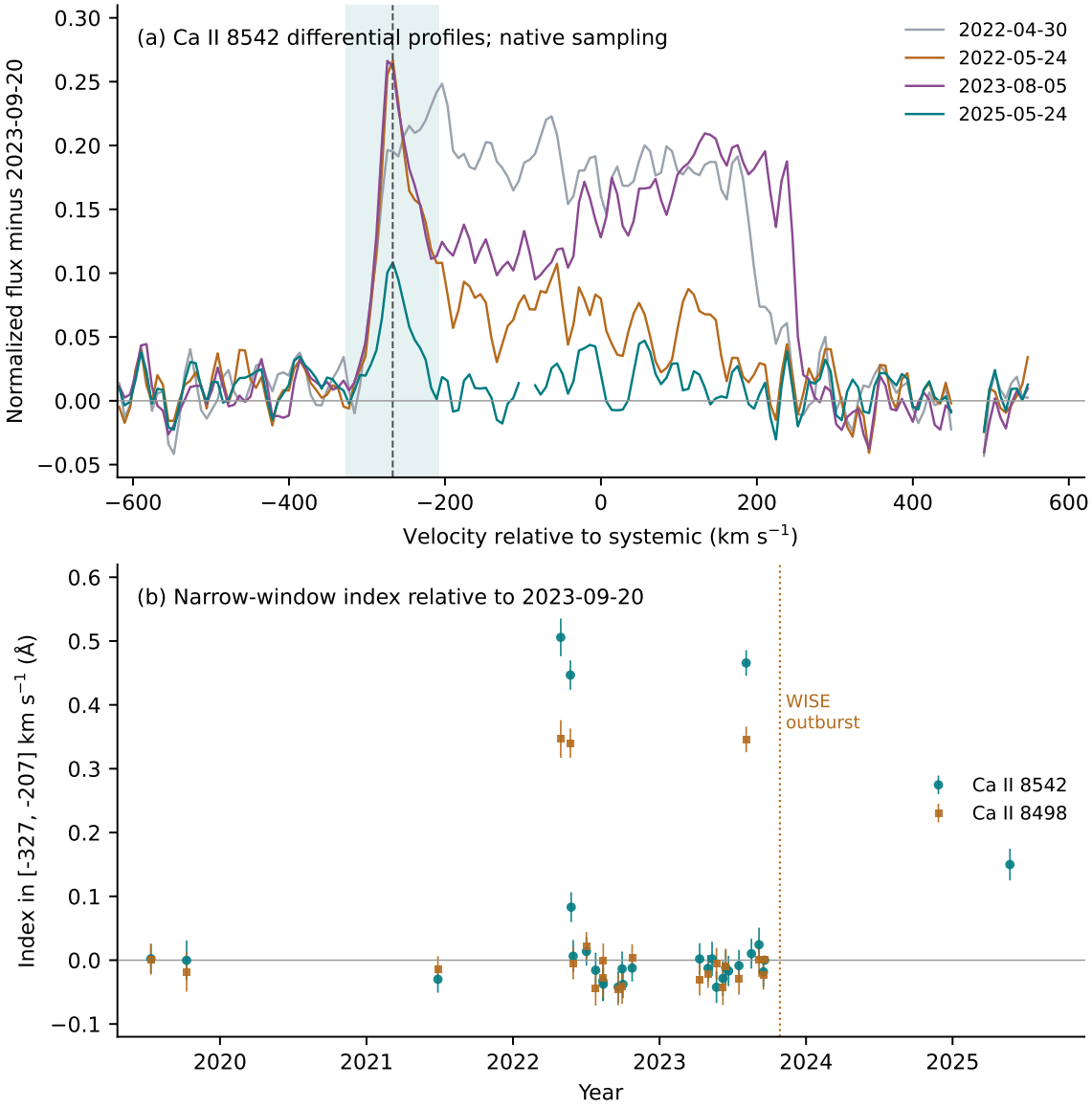


Figure 1: (a) Ca II 8542 normalized flux of the archive products minus the quiet 2023 September 20 spectrum for the four enhanced nights, unsmoothed; the dashed line marks −263 km s^{-1} and the band the narrow window. Blank segments are pipeline-flagged pixels in either spectrum (reference: +456 to +484 km s^{-1}; 2025: near −95 km s^{-1}; all: beyond +550 km s^{-1}) and the 8548.0 Å sky residual (+239 km s^{-1}). (b) Narrow-window index for all 29 nights at 8542 Å (circles) and 8498 Å (squares; absent in 2025 because that archive product is flagged there); open symbols mark 2022 April 30, where the excess is the broad profile. Error bars are the per-night line-free scatter of the same integral.